ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION 2024-25

B.Tech. CSE (AI)

Semester: VI

SUBJECT CODE: BCSAI604

NATURAL LANGUAGE PROCESSING

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (10 Marks)

Q. 1. Answer the following questions: (1 × 5 = 5 Marks)

(i) Define regular expressions in NLP.
(ii) What is morphology in NLP?
(iii) What is n-gram models?
(iv) What is entropy in language modeling?
(v) What is part-of-speech (POS) tagging?

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) Define a parse tree.
(ii) What is a context-free grammar (CFG)?
(iii) What is semantics in NLP?
(iv) Name three real-world applications of NLP.
(v) What is machine translation?

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) What is the role of n-grams (unigrams, bigrams, trigrams) in the Bag of Words model?

OR

(b) What are stopwords in NLP, and why are they typically removed during text preprocessing?

(ii)(a) Write about different smoothing techniques in n-gram models.

OR

(b) Differentiate between top-down parsing, bottom-up parsing.

Q.4.(i)(a) Define discourse analysis and explain its significance in NLP applications.

OR

(b) What is tokenization in NLP? Discuss word-level tokenization and provide example.

(ii)(a) Write about different phases of Natural Language Processing (NLP).

OR

(b) What is semantic analysis in Natural Language Processing (NLP), and how does it differ from syntactic analysis?

SECTION – C

NOTE: Attempt all the questions. (10 × 2 = 20 Marks)

Q.5. (a) Explain different approaches used for WSD, including supervised, unsupervised, and knowledge-based methods**.**

OR

(b) What is POS tagging? Explain different approaches used in POS tagging.

Q.6. (a) Explain the process of stemming in NLP. Discuss different types of stemmers in detail.

OR

(b) Explain the different approaches to MT (Machine translation), including Rule-Based Machine Translation (RBMT), Statistical Machine Translation (SMT), and Neural Machine Translation (NMT)**.**